

CONNECTIONS

1 INTRODUCTION

Sometimes the required controls can be so simple that using a programmable controller does not make sense. In this task, you will practice implementing an application using only relays, buttons and sensors to control actuators in the desired way. The purpose is to practice reading circuit diagrams and making connections accordingly. In addition, you will become familiar with some common pneumatic components during the work.

2 INFORMATION ABOUT THE WORKSTATION

The work is performed using a Festo training table, which has the components used in the work already attached.

3 SAFETY

There is a risk of crushing in the workplace. Ask the supervisor to be present before applying pressure to the system. Always turn off the pressure during repair work, etc., to avoid damage. NOTE! The cylinder rod may start moving as soon as you apply pressure to the system. Therefore, never put your hands, fingers, or other body parts in the operating area of the cylinder when the equipment is pressurized.

4 TASK

The task is to implement the control of a cylinder electro pneumatically without programmable logic. The application starts with the spring-return pushbutton SET and stops by releasing the pushbutton RESET. While the application is running, the cylin-

CONNECTIONS

der piston makes a continuous back and forth movement between the inner and outer positions. The outer and inner positions are detected by an electromechanical switch. There are indicator lights for the inner and outer control, which show the control status. When the basic application is working, present it to the operator. Finally, examine the operation of the inductive sensor. Test the operation by connecting it electrically, but leave it mechanically disconnected, which allows you to move the sensor against different materials. Electrical connection means that the sensor must receive operating voltage (red) and operating ground (blue). The sensor supplies voltage to the black wire when activated (when detecting metal). Connect the sensor to the panel so that you can see the indicator light when a metal object is present. After testing the sensor with a few different materials, also connect it mechanically so that you can use it to detect the external position of the cylinder piston. Adjust the sensor to a suitable distance from the piston so that identification is successful. Find the maximum distance at which identification still occurs reliably. Present your work and results to the instructor. Be prepared to answer questions about it.

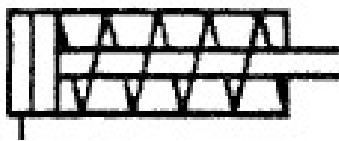
Tips: You can take the supply voltage for your circuits from the push-button unit (+ at the top and – at the bottom). The attached picture shows the principles of the electrical and pneumatic connection of the circuit.

5 ABOUT PNEUMATIC COMPONENTS

The tasks require a bi-stable 5/2-way valve and a double-acting cylinder. The workstation may also have a mono-stable 3/2-way valve, a single-acting cylinder, a resistance check valve, etc. components that are not necessarily needed to perform the tasks.

Single-acting cylinder (spring-return)

A single-acting cylinder works in such a way that when a sufficiently high pressure is connected to the compressed air port, the piston and piston rod move to the outer position, i.e. the + position. When the pressure in the compressed air port drops sufficiently low, a return spring built into the cylinder pushes the piston and piston rod back to the inner position, i.e. the – position. A single-acting cylinder is not used in this exercise.

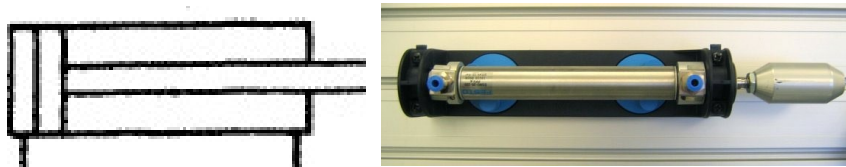


Double-acting cylinder (compressed air drives in both directions)

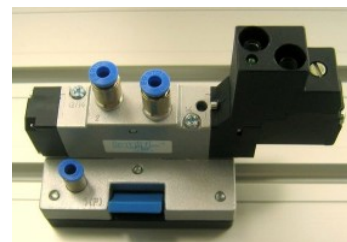
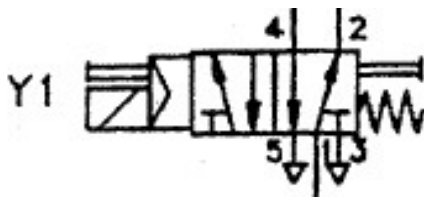
A double-acting cylinder differs from a single-acting cylinder in that it does not have a built-in return spring, but the piston and piston rod are returned to the internal, or –

CONNECTIONS

position, by air pressure. So, as the picture shows, a double-acting cylinder has two compressed air ports.

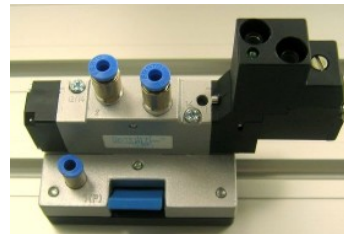
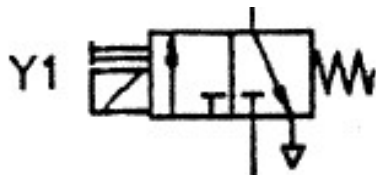
**5/2-way valve, monostable (electrically controlled, spring-returned)**

The name of the 5/2-way valve comes from the fact that it has 5 compressed air ports and 2 positions. Monostable means that the valve remains in only one state without external control, i.e. it is controlled electrically and when the voltage is switched off, the spring returns the valve to its basic state. As can be seen from the diagram below, in the idle state, air flows from port 1 of the valve to port 2 and from port 4 to port 5, which is the pressure relief port (directly open to the outside air). When voltage is applied to the valve coil (= the valve is controlled to turn on), you can think of the valve element in the diagram as moving one notch to the right while the air pipes connected to it remain in place. In this case, the air flows from port 1 to port 4 and from port 2 to port 3, which in this valve, like port 5, is an exhaust port (a port that releases pressure to the room air). As can be seen from the pictures, only three compressed air pipes are connected to the valve (to ports 1, 2 and 4). The other two ports (5 and 3) are led to the room air through internal bores in the valve.

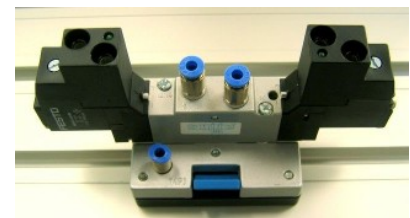


CONNECTIONS**3/2-way valve, monostable (electrically controlled, spring-return)**

The operating principle of a 3/2-way valve is similar to that of a 5/2-way valve, but it only has 3 ports. This valve type also has 2 positions. A 3/2-way valve is not used in this exercise.

**5/2-way valve, bistable (electric-electric control)**

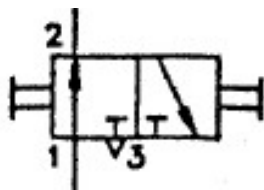
A bistable 5/2-way valve differs from a monostable 5/2-way valve in that it is not spring-returned. The valve has two different control coils and the valve remains in the position it was last controlled to. One could think that control coil Y1 tends to control the valve element one notch to the right and control coil Y2 one notch to the left.

**3/2-way valve, manually operated**

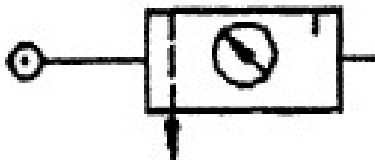
The device works in the same way as the previously presented electrically controlled 3/2-way valve, except for the control method. As its name suggests, this valve is con-

CONNECTIONS

trolled manually by pressing a green button, which moves the valve element mechanically. The drawing shown below does not correspond to the device in the system, as the device used is spring-return and does not allow air to pass through in the idle state.

**Service device**

Air is supplied to the system through the service device. In the training equipment, the service device includes a filter, an adjustable pressure reducer, a pressure gauge and a finger-operated shut-off valve (the red knob in the picture).



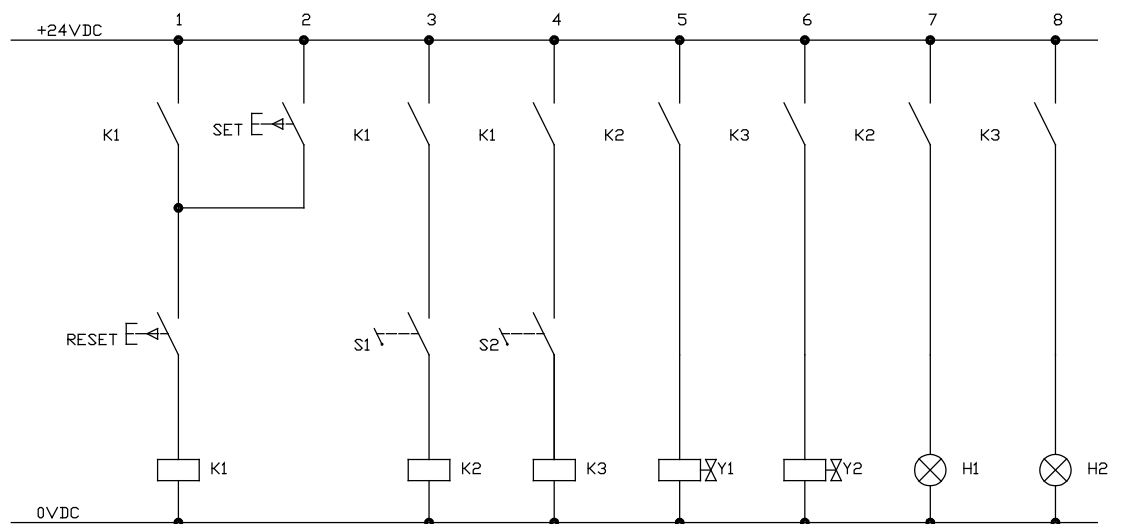
CONNECTIONS

FIGURE 1 System circuit diagram

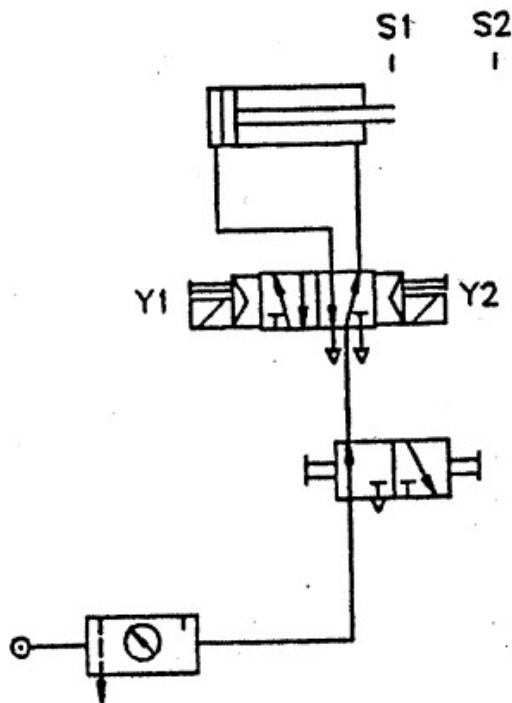


Figure 2 System pneumatic diagram